

injection mechanism make use of computers these days. The optimal fuel injection levels for all possible conditions that a car can come across are programmed into these ECUs.

Now, if you are wondering what the big deal is all about, here it is. Remember all those cold mornings when you struggled with getting your car started? You turned the key—zilch, nothing happened! This is because it is difficult to get fuel to burn in cold conditions. In old cars, which do not make use of ECUs for engine management, it takes a while before you can coax the engine to start. But, if your car makes use of an ECU, the computer is fed with the temperature of the engine coolant and it detects that the igniting spark needs to be on for a longer time. The ECU then generates the required sparking sequence. That done, the engine starts up faster than it would have on a car without an ECU for engine management.

Computers are also used in emission control and better fuel consumption. In fact, it was the problem of emissions rather than improved performance that first led to the introduction of computers for engine control and management. Sensor inputs of parameters such as oxygen, engine temperature, air pressure, air temperature, and throttle position are continuously analysed by the computer. This analysis helps it control fuel injectors, spark plugs and idling speed to get the best performance and low levels of emission. Some cars even incorporate computers that regulate the octane content in fuel. This is very useful in a country like India where fuel adulteration is not uncommon.

So what do you say about a car like Honda City? Cute, huh? But this compact car has a brainy ECU to control its engine valves. The City makes use of VTEC (Variable Timing and Engine Control) technology for optimum engine performance and better fuel efficiency. At various engine speeds, this ECU modifies valve operation for the best possible performance and programmed fuel injection makes sure that fuel is used in the best possible manner.

Cars with ESP?

No, we are not talking about extrasensory perception here. ESP in 'car talk' stands for Electronic Stability Program. The advantage of ESP

is that it helps correct driving errors such as excessive acceleration or taking corners at dangerously high speeds. In many cases ESP makes use of Antilock Brake Systems (ABS) too (see box, 'Cruising to Safety' for an insight into ABS). An ESP monitors and compares the direction of vehicle travel with the driver's actions. These actions take place around 25 times per second. The ESP sensors recognise any signs of instability, understeer or oversteer as soon as it happens. It then proceeds to stabilise the vehicle using braking on each individual wheel along with a bit of engine management.

If you are understeering a car that is without ESP, it loses traction and begins to slide. But if your car is equipped with an ESP system, the latter immediately detects this—brakes are applied to the inner rear wheel and engine power is

reduced as well. This helps ensure that your vehicle does not go into an uncontrollable skid.

If you are oversteering, the rear wheels lose their grip and begin to skid. The ESP then applies brakes to the outer front wheel, which helps to bring the vehicle back into control.

There's more to come

The uses we have just seen are not all. Besides the purely utilitarian functions that car computers serve, they can and are used for many other applications. Even the CD player or radio in your car makes use of a computer, usually the simplest kind. This computer controls functions such as remembering the CD sequence if you have a CD changer, or remembering the preset stations of the radio.

The security system of your car (if it's the key-less type) is also controlled by nothing less than a computer. But that's not all. There are several models of cars these days that come with self-levelling suspensions. If you are using such a model, you'll find that your car adjusts its suspension settings to suit the kind of road on which it's travelling. Once again, it's computers connected to sensors which are behind these suspensions. If several people use your car, you can set and save mirror and seat adjustments to suit each individual user—all by the means of a computer. As people demand more from their cars, the use of computers in cars is going to increase. All this to make driving a safe and pleasant experience even for those who consider driving a chore. 

Cruising to Safety

ABS (Antilock Brake Systems) is a feature that has been around for a long time. The main advantage of this system is that it prevents the wheels from locking while you are braking. If the wheels lock, the car could go into an uncontrollable skid resulting in a crash. Cars having ABS have optimum vehicle stability. Braking distance is also lesser in a car with ABS as compared to one that is without it. Needless to say, this is a boon in a country like India, where road conditions can be quite awful.

A typical ABS arrangement consists of sensors located in all four wheels and an ABS ECU (Electronic Control Unit). These sensors send the status of a wheel that is slowing down too fast to the ECU. The ABS ECU checks the status and then reduces the air pressure of the brake to that wheel which means the braking on that wheel is released. This ensures that the wheel does not lock up and does not go into a skid. Almost all ABSs have facilities for self-checking when the car starts to move. If the system is not working properly, the ABS is switched off and the conventional braking system comes into effect.

Computing technology also helps in cruise control, which keeps you at a safe distance from the vehicle in front of you. These systems make use of radar units to keep track of the distance and relative speed of any vehicle in your car's path. The distance and speed values are then used by the computer to control the throttle and brakes.

